

Day: 13

* What is NetworkMiner?

- NetworkMiner is a passive network forensics tool — it reads a PCAP and automatically extracts everything useful without you having to apply filters manually. Real analysts use it for rapid triage — get the big picture fast.

* NetworkMiner vs Wireshark

Feature	Wireshark	NetworkMiner
Primary use	Deep packet inspection.	Rapid triage and extraction
Learning curve	Steep	Easy
Shows hosts	Manual (Statistics menu)	Automatic (Hosts tab)
Extracts files	Manual (Export Objects)	Automatic (Files tab)
Extracts credentials	Manual (Follow stream)	Automatic (Credentials tab)
Best For	Detailed Analysis	First 5 minutes of investigation
Speed	Slow (Manual)	Fast (Automatic)

* NetworkMiner Tabs — What Each Shows

Tab	What it shows	DFIR Use
Hosts	All IPs, MAC addresses, OS fingerprint	Who was on the network?
Files	Every file transferred	Malware dropped?
Credentials	Username and passwords	Were credentials stolen?
DNS	All DNS queries and responses	C2 domains?
Images	Images transferred	What was exfiltrated
Parameters	HTTP POST parameters	Form data, Stolen info

* Wireshark Statistics Menu — Key Features

1) Protocol Hierarchy

Statistics → Protocol Hierarchy

— Shows every protocol as a percentage of total traffic.

— What to look for:

- Unexpected protocols C.e.g. IRC, Telnet)
= red flag

- Very high DNS % = possible DNS tunneling

- Very high ICMP % = possible ICMP tunneling or ping sweep.

2) Conversations

Statistics → Conversations → TCP tab

— Shows which hosts talked the most and how much data was transferred

— What to look for:

- External IP with huge data transfer = exfiltration

- Many connections to single IP = C2 beaconing

3) IO Graph

Statistics → IO Graph

— Shows traffic volume over time as a graph.

— What to look for:

- Regular spikes every X seconds =

beaconing.

- Sudden massive spike = data exfiltration burst.

4) DNS Statistics

Statistics → DNS

— Shows all DNS query types and response codes

— What to look for:

- High NXDOMAIN (no such name) count = DGA malware failing to find C2.
- Many queries to same domain = beaconing.

* The Analyst Triage Workflow

STEP 1: Open in NetworkMiner (2 minutes)
→ Get host list, check Files and Credentials tabs.

STEP 2: Protocol Hierarchy
→ Any unexpected protocols?

STEP 3: Conversations
→ Who is talking most? External IPs?

STEP 4: IO Graph
→ Any beaconing pattern?

STEP 5: DNS filter in Wireshark
→ Suspicious domains?

STEP 6: HTTP filter
→ Suspicious POST requests?

STEP 7: Follow TCP Streams on suspicious connections
→ Read the actual conversation

* Ethernet Frames and MAC Addresses (Book topic)

Field	Size	Purpose
Destination MAC	6 bytes	Who receives this frame
Source MAC	6 bytes	Who sent this frame
Type	2 bytes	IPv4 = 0x0800, ARP = 0x0806
Data (payload)	46-1500 bytes	The actual IP packet
CRC	4 bytes	Error Detection

* Key Fact: Mac addresses are Layer 2 — they only exist within a Local Network. Every router hop replaces the MAC address but keeps the IP address.

* Diagrams:

- Analyst triage workflow - always this order

1) NM
Fast Triage



2) Proto
Hierarchy



3) Conv
Who talks?



4) IO
Graph



5) DNS
C2 Domains



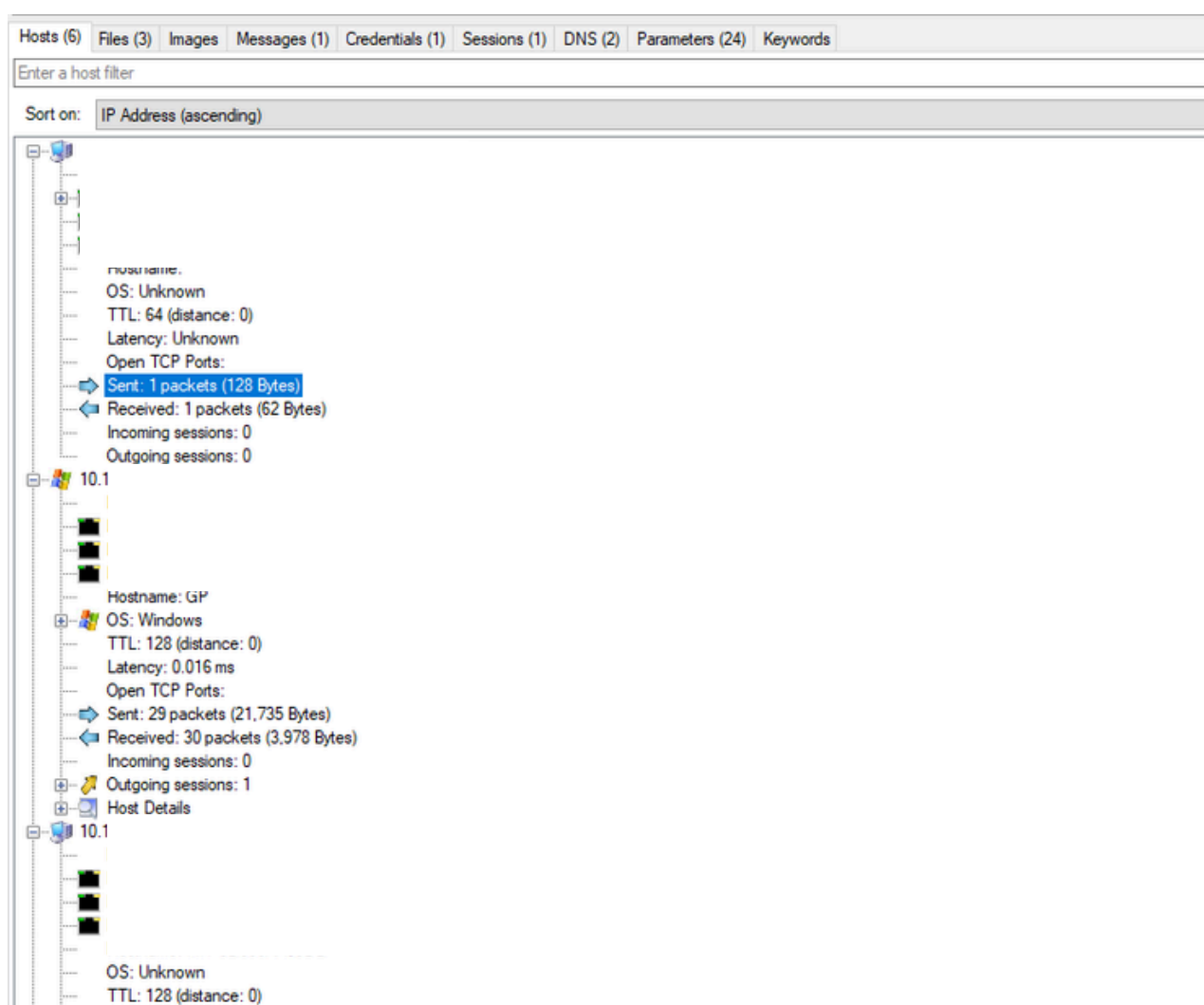
6) HTTP
POST = enfil?

- NetworkMiner : Automatic extraction (first 5 minutes)
- Wireshark : Deep manual analysis (The Details)

Part 1 — NetworkMiner:

1. Download and install NetworkMiner free from netresec.com
2. Download Open a malware PCAP
3. NetworkMiner automatically extracts:

- Hosts tab — all hosts discovered
- DNS tab — all DNS queries
- Files tab — all files transferred
- Credentials tab — any credentials found
- Images tab — any images transferred



Part 2 — Wireshark Statistics:

1. Open the same PCAP in Wireshark
2. Go to Statistics → Protocol Hierarchy and observe
3. Go to Statistics → Conversations → TCP tab---->Which IP has the most traffic?
4. Go to Statistics → DNS----->What domains were queried most?
5. Go to Statistics → HTTP → Requests----->What URLs were requested?
6. Go to Statistics → IO Graph----->Watch traffic spikes — beaconing shows as regular spikes!

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
▼ Frame	100.0	60	100.0	26866	23 k	0	0	0	60
▼ Ethernet	100.0	60	3.4	924	803	0	0	0	60
▼ Internet Protocol Version 4	100.0	60	4.5	1200	1043	0	0	0	60
▼ User Datagram Protocol	5.0	3	0.1	24	20	0	0	0	3
▼ NetBIOS Datagram Service	1.7	1	0.3	82	71	0	0	0	1
▼ SMB (Server Message Block Protocol)	1.7	1	0.4	119	103	0	0	0	1
▼ SMB MailSlot Protocol	1.7	1	0.1	25	21	0	0	0	1
Microsoft Windows Browser Protocol	1.7	1	0.1	33	28	1	33	28	1
Domain Name System	3.3	2	0.5	134	116	2	134	116	2
▼ Transmission Control Protocol	88.3	53	4.0	1076	935	21	436	379	53
▼ Simple Mail Transfer Protocol	53.3	32	73.1	19631	17 k	31	19602	17 k	32
Internet Message Format	1.7	1	78.0	20966	18 k	1	20966	18 k	1
Internet Control Message Protocol	6.7	4	8.3	2224	1934	4	2224	1934	4

Wireshark · Conversations · smtp.pcap

Conversation Settings

☐ Name resolution

☐ Absolute start time

Ethernet · 2		IPv4 · 4		IPv6		TCP · 1		UDP · 2							
Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A	Flows
10.10.1.4	1470	74.53.140.153	25	53	24 kB	0	28	22 kB	25	2 kB	0.036986	7.5777	23 kbps	2090 bits/s	19

Wireshark · DNS · smtp.pcap

Packet Type	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
Total Packets	2				0.0588	100%	0.0200	0.000
▼ Answer Type	4				0.1176	100%	0.0400	0.000
NS	2				0.0588	50.00%	0.0200	0.000
CNAME	1				0.0294	25.00%	0.0100	0.000
A	1				0.0294	25.00%	0.0100	0.000
▼ rcode	2				0.0588	100%	0.0200	0.000
No error	2				0.0588	100.00%	0.0200	0.000
▼ opcodes	2				0.0588	100%	0.0200	0.000
Standard query	2				0.0588	100.00%	0.0200	0.000
▼ Response	2				0.0588	100%	0.0200	0.000
Response	1				0.0294	50.00%	0.0100	0.000
Query	1				0.0294	50.00%	0.0100	0.000

Request Type	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
HTTP Requests by HTTP Host 0						100%	-	-

